

TABLE 1 Primers used for amplification of the complete genome

Primer name ^a	Sequence (5'–3')	Amplified length (bp)	Nucleotide position range	<i>T_m</i> ^b (°C)
PDCoV-1-F	ACATGGGGACTAAAGATAAAAATTATAGC	3,300	1–3300	58
PDCoV-3300-R	GCTCATCGCCTACATCAGTG	3,300	1–3300	58
PDCoV-3091-F	CGGATTTAAAACACAGACT	1,770	3091–4860	51
PDCoV-4860-R	ACGACTTTACGAGGATGAAT	1,770	3091–4860	51
PDCoV-4741-F	CTCCTGTACAGGCCTTACAA	1,680	4741–6420	55
PDCoV-6420-R	TCACACGTATAGCCTGCTGA	1,680	4741–6420	55
PDCoV-6291-F	CTCAATGCAGAAGACCAGTC	1,751	6291–8041	53
PDCoV-8041-R	CAGCTTGGTCTTAAGACTCT	1,751	6291–8041	53
PDCoV-7920-F	GGTACTGCTTCTGATAAGGAT	1,741	7920–9660	53
PDCoV-9660-R	TAGGTACAGTTGTGAACCGA	1,741	7920–9660	53
PDCoV-9463-F	TACTCTTCACAGCCTTCAC	1,668	9463–11130	51
PDCoV-11130-R	GCAAATCCAGGACCCATAGTAG	1,668	9463–11130	51
PDCoV-10560-F	CGCTACACTATTGTGAAGAA	2,288	10560–12847	52
PDCoV-12847-R	TTCGTAGGGCTCAAGATA	2,288	10560–12847	52
PDCoV-12301-F	TCCAGATGACCGTTGCGTAT	2,652	12301–14952	55
PDCoV-14952-R	CCAACAGAGTCGGGTAAG	2,652	12301–14952	55
PDCoV-14703-F	CACCATAACGAAGAACC	2,483	14703–17185	54
PDCoV-17185-R	CCGATGAGTGTCGTAGCG	2,483	14703–17185	54
PDCoV-17165-F	TGCCGCTACGACACTCAT	2,039	17165–19203	56
PDCoV-19203-R	TCCGCTAAAGGAGAATAGTTG	2,039	17165–19203	56
PDCoV-18485-F	TGCTACCCAATCTTACAGT	2,595	18485–21079	53
PDCoV-21079-R	GCAAATACTCCGTCTGAAC	2,595	18485–21079	53
PDCoV-20761-F	GTCTTACCGTGTGAACCCC	3,065	20761–23825	55
PDCoV-23825-R	AACCACGAGACTGTAAGCAA	3,065	20761–23825	55
PDCoV-23804-F	TTTTGCTTACAGTCTCGTGGT	1,616	23804–25419	55
Oligo(dT)	TTTTTTTTTTTTTTTTT	1,616	23804–25419	55

^a Each primer was designed based on the HKU15-155 strain (GenBank accession number [JQ065043](#)).

^b *T_m*, melting temperature.

genomic structure of the two PDCoVs contained the following gene order: 5' untranslated region (UTR), open reading frame 1a/1b (ORF 1a/1b), spike glycoprotein (S), envelope (E), membrane (M), nonstructural protein 6 (Nsp6), nucleocapsid (N), Nsp7, and 3' UTR. The sizes of these genes of the PDCoV strain CH-01 were 539, 18,788, 3,480, 252, 654, 285, 1,029, 603, and 392 nt, respectively. The PDCoV strain HNZK-02 contained an 18,803-nt ORF1a/1b, which had a 6-nt and 9-nt insert at nucleotide positions 1739 and 2804, respectively, compared with that of the CH-01 strain (the nucleotide positions were numbered according to the entire genome sequence of PDCoV CH-01). The other genes in strain HNZK-02 had the same sizes as those of the PDCoV CH-01 strain.

The identities of PDCoV nucleotide and deduced amino acid sequences were aligned and analyzed using the Clustal W program of DNASTar 7.0 green. The two sequenced PDCoV strains have nucleotide identities from 98.1% to 99.4% with the Chinese epidemic PDCoV strains (GenBank accession numbers [KP757892](#), [KR131621](#), [MF280390](#), [KU981062](#), and [KP757890](#)) and nucleotide identities from 98.9% to 99.0% with some U.S. PDCoV strains (GenBank accession numbers [KJ481931](#), [KJ462462](#), [KJ567050](#), and [KJ769231](#)).

The sequence data of the two PDCoV strains will facilitate future research on the epidemiology and evolutionary biology of PDCoV in China.

Data availability. The complete genome sequences of the two PDCoV strains (HNZK-02 and CH-01) have been deposited in GenBank under the accession numbers [MH708123](#) and [KX443143](#).

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